REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Acceleration Practice Questions

Grade 9 Advanced | Module 3 Lesson 1 | Acceleration

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: aavg = (vf − vi)/Δt; velocity–time slope gives acceleration; area under a v–t graph gives displacement.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 54–62.

P01. Acceleration describes the rate of change of:

A. Position only B. Velocity
C. Mass D. Time

P02. The SI unit of acceleration is:

A. m/s B. m^2/s
C. m/s^2 D. s/m

P03. A positive velocity and negative acceleration indicate:

A. Speeding up necessarily B. Negative position necessarily
C. Rest forever D. Slowing down at that instant

P04. A horizontal velocity–time line indicates:

A. Zero acceleration B. Zero velocity necessarily
C. Increasing acceleration D. A changing displacement rate

P05. The signed area under a velocity–time graph gives:

A. Acceleration B. Displacement
C. Mass D. Average force

P06. Velocity changes from +4 m/s to +16 m/s in 6 s. Calculate average acceleration.

A. 4 m/s^2 B. 1 m/s^2
C. 2 m/s^2 D. -2 m/s^2

P07. A car slows uniformly from +12 m/s to rest in 4 s. Calculate its acceleration.

A. -6 m/s^2 B. -1.5 m/s^2
C. 3 m/s^2 D. -3 m/s^2

P08. A velocity–time graph is horizontal at +6 m/s for 8 s. Calculate displacement from the graph.

A. 48 m B. 96 m
C. 24 m D. -48 m

P09. A straight velocity–time graph rises from 0 to +12 m/s in 4 s. Calculate the displacement.

A. 48 m B. 24 m
C. 12 m D. -24 m

P10. An object has initial velocity +4 m/s and constant acceleration +3 m/s² for 3 s. Find its final velocity.

A. 26 m/s B. 6.5 m/s
C. 13 m/s D. -13 m/s

P11. Define average acceleration.

P12. Can negative acceleration make an object speed up?

P13. What does the slope of a velocity–time graph represent?

P14. Distinguish velocity from acceleration.

P15. Can an object have zero velocity and nonzero acceleration at an instant?

P16. Velocity changes from +6 m/s to +24 m/s in 9 s. Calculate average acceleration.

P17. Velocity changes from +8 m/s to +32 m/s in 12 s. Calculate average acceleration.

P18. A car slows uniformly from +18 m/s to rest in 6 s. Calculate its acceleration.

P19. A car slows uniformly from +24 m/s to rest in 8 s. Calculate its acceleration.

P20. A velocity–time graph is horizontal at +9 m/s for 12 s. Calculate displacement from the graph.

P21. A velocity–time graph is horizontal at +12 m/s for 16 s. Calculate displacement from the graph.

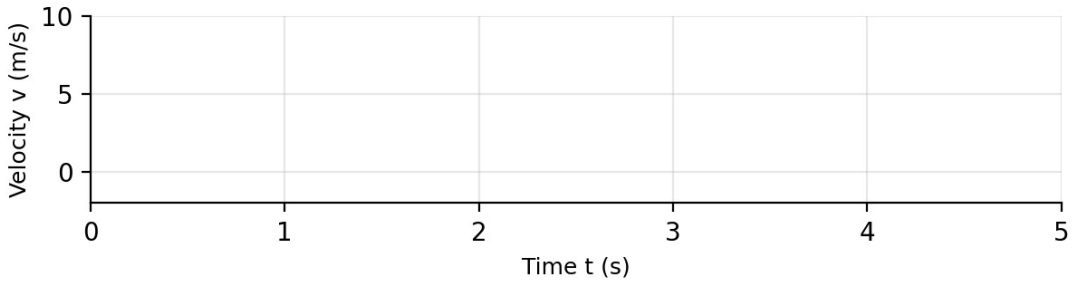
P22. A straight velocity–time graph rises from 0 to +18 m/s in 6 s. Calculate the displacement.

P23. A straight velocity–time graph rises from 0 to +24 m/s in 8 s. Calculate the displacement.

P24. An object has initial velocity +6 m/s and constant acceleration +3 m/s² for 4 s. Find its final velocity.

P25. An object has initial velocity +8 m/s and constant acceleration +3 m/s² for 5 s. Find its final velocity.

P26. Plot t = 0, 1, 2, 3, 4 s against v = 2, 4, 6, 8, 10 m/s. Find acceleration and displacement in 4 s.



P27. An object has v = −6 m/s and a = +2 m/s². Is it speeding up or slowing down initially? Explain.

P28. A student says negative acceleration always means slowing down. Give a counterexample.

P29. A v–t graph lies below the time axis. Explain the sign of its area and distinguish displacement from distance.

P30. Describe two ways that velocity can change.